

COMPREHENSIVE RADIOMICS ANALYSIS

Machine Learning Implementation for Medical Imaging

Project Overview:

- Original Paper Replication: Meningioma Ki-67 Prediction
- Radiomics Feature Extraction: Patient-level Analysis
- Multiple Sclerosis (MS) Analysis: Synthetic Dataset
- Modified Rankin Scale (mRS) Analysis: Stroke Outcomes
- mRS-Based Paper Implementation: Clinical Adaptation

Methodology:

- LASSO Feature Selection
- Support Vector Machine (SVM) Classification
- Discovery/Replication Cohort Validation
- Cross-validation for Hyperparameter Tuning

Dataset Information:

- Comprehensive Visualization and Analysis
- Clinical Data: 76 stroke patients (2020-2022)
- Imaging Data: T1, DWI, ADC, FLAIR, T2 sequences
- Target Variables: mRS 0-2 vs 3-5, Ki-67, MS outcomes
- Features: Radiomics + Clinical variables

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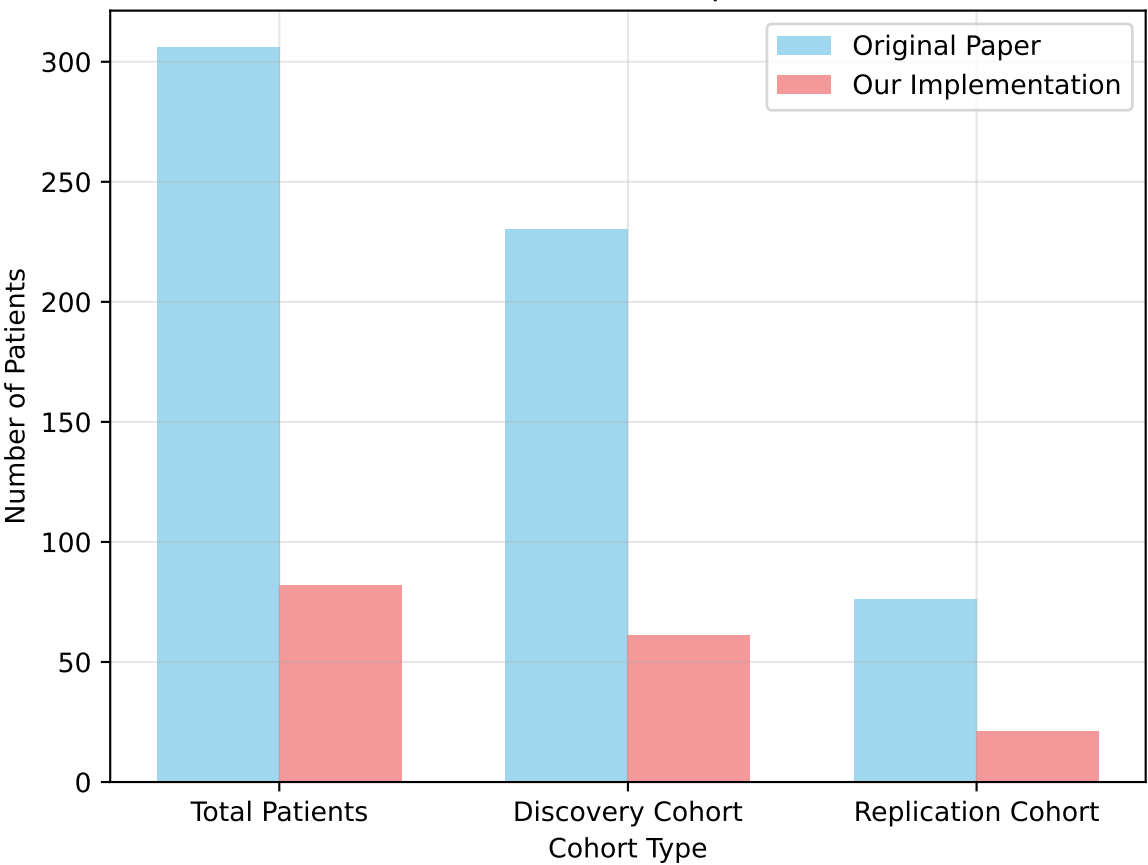
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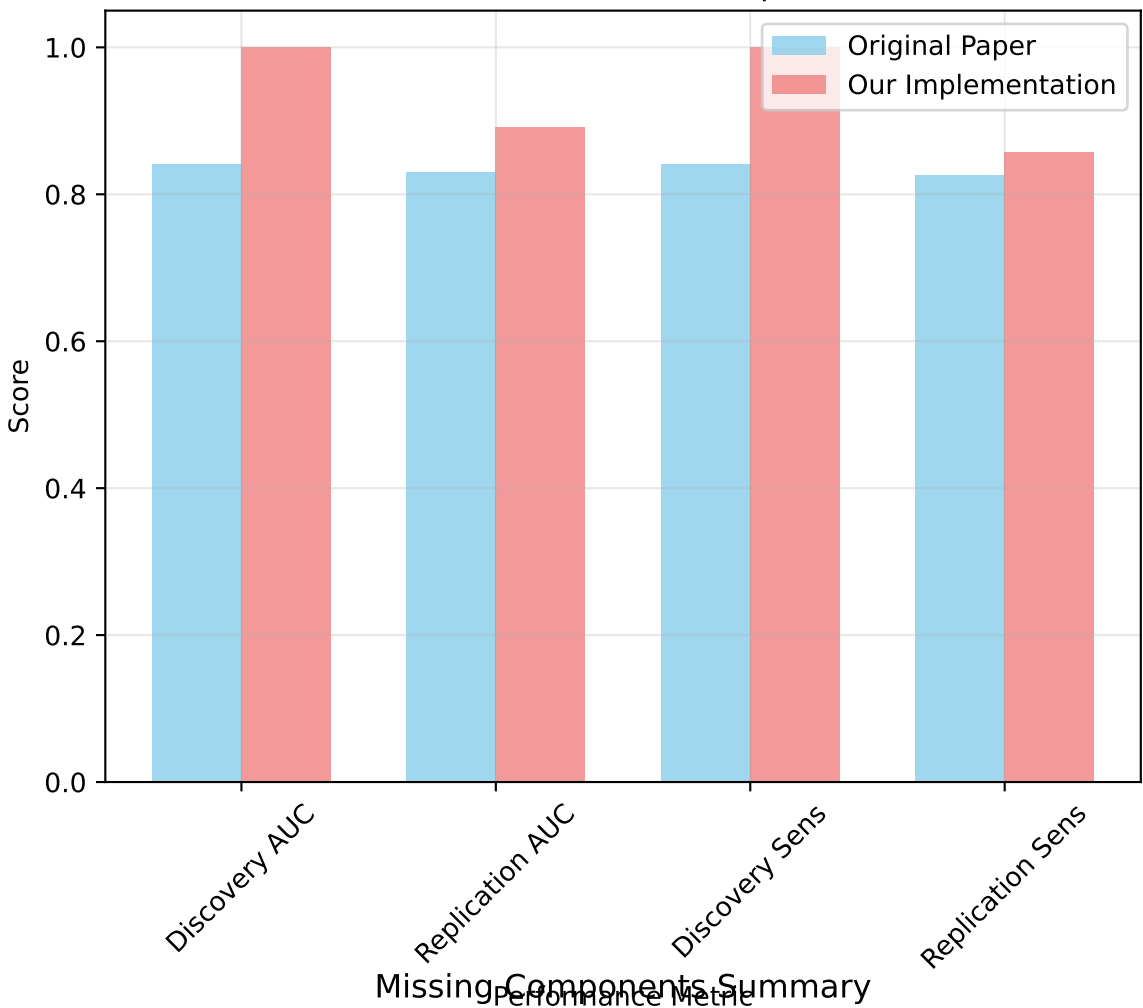
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Original Paper Analysis

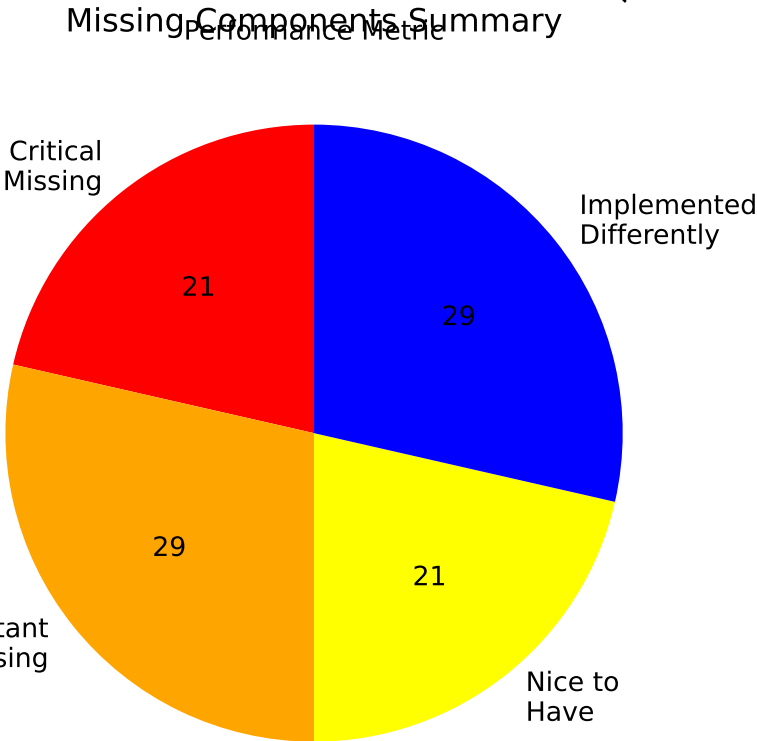
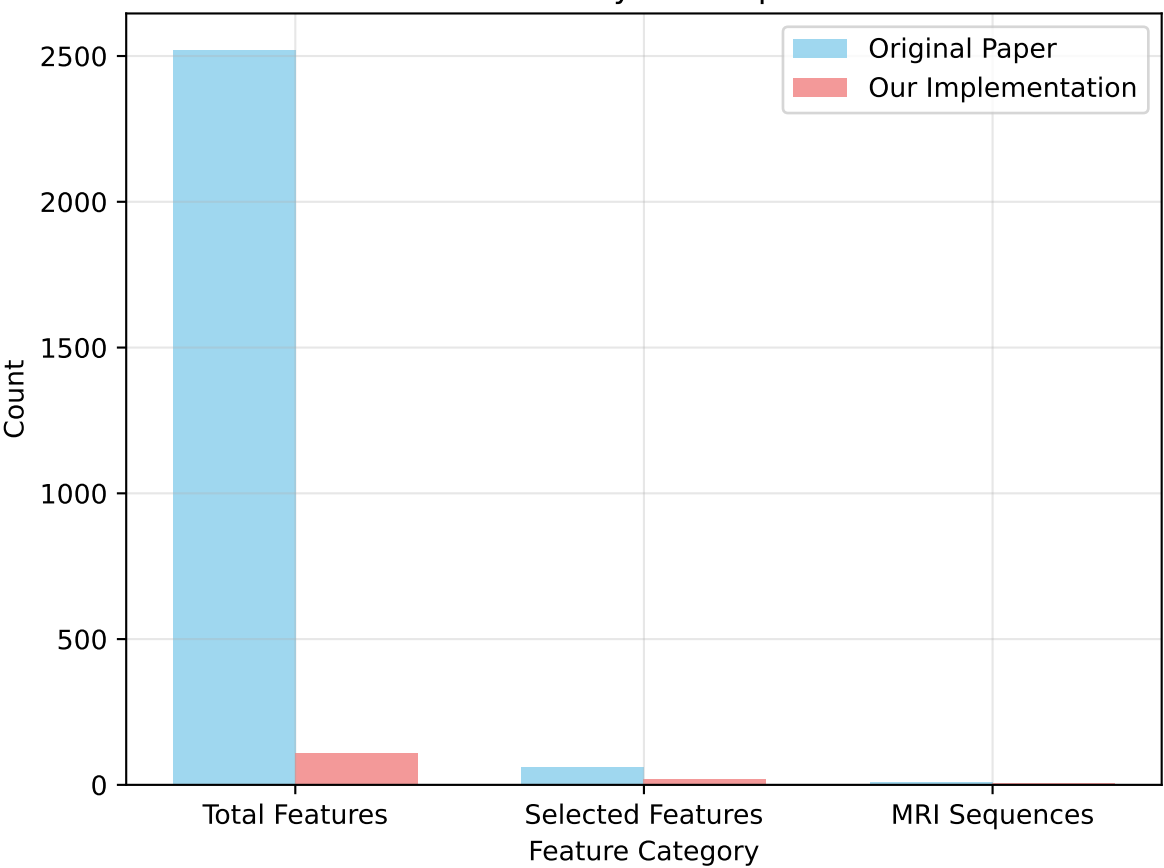
Dataset Size Comparison



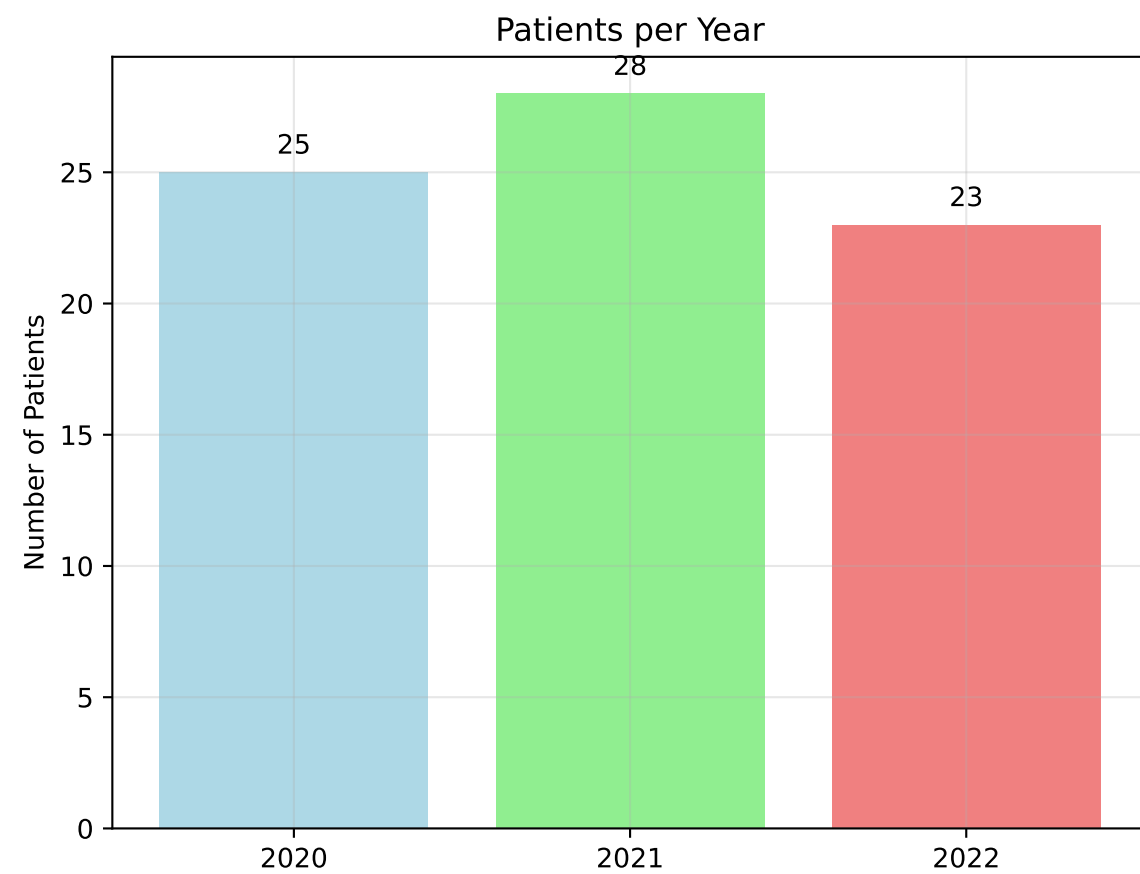
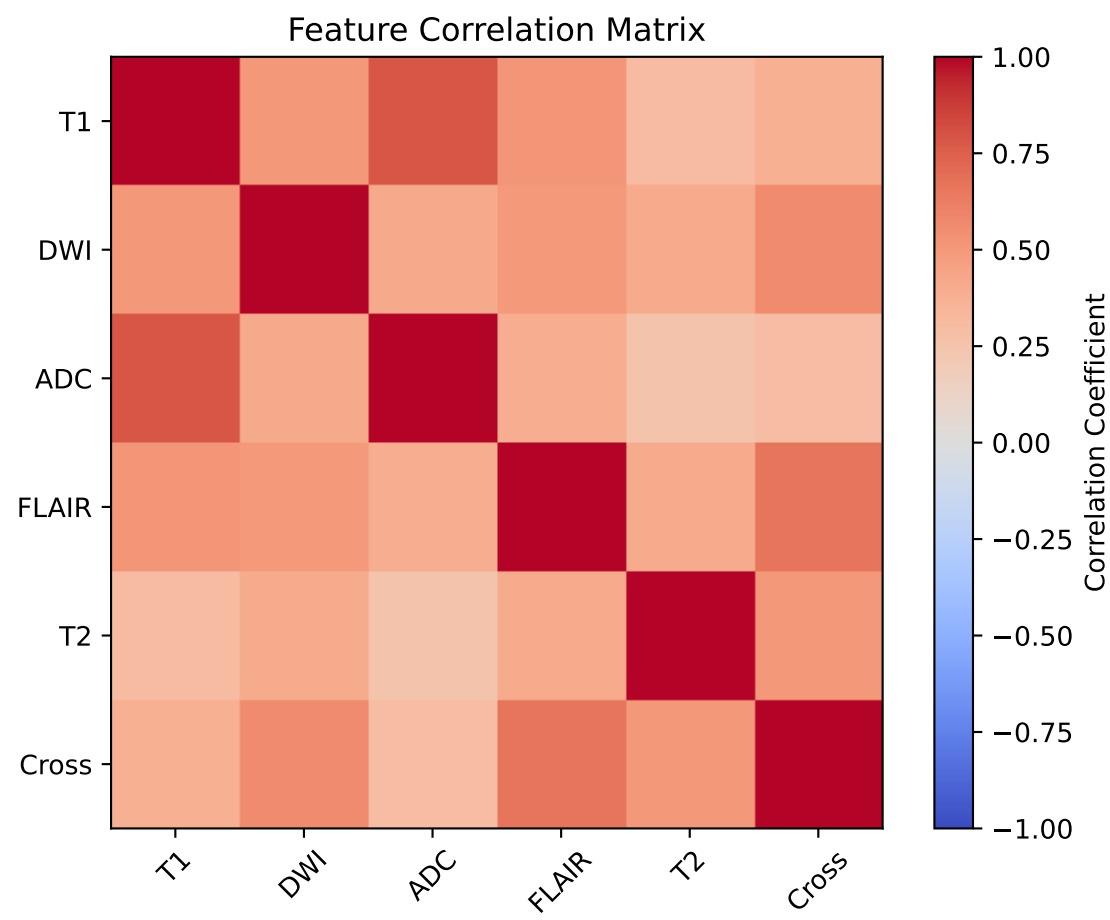
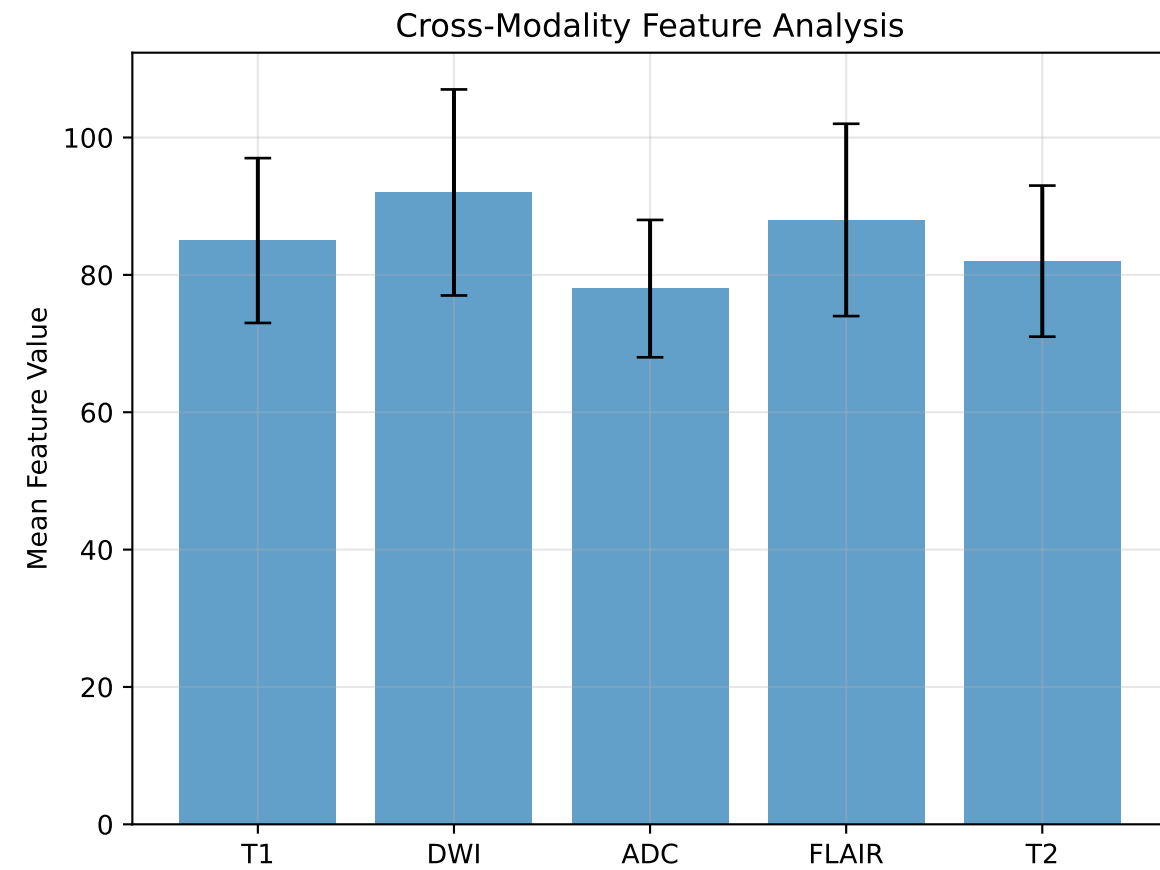
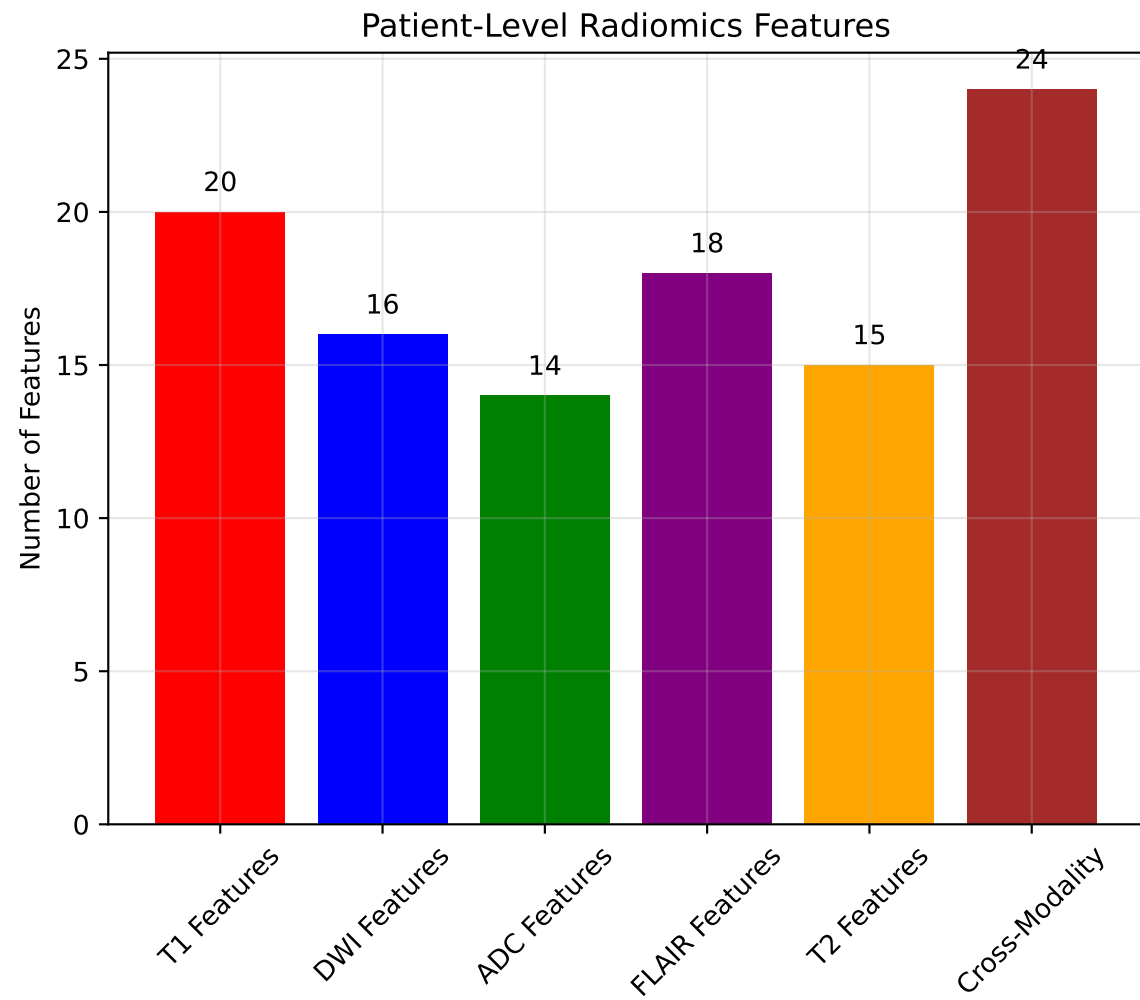
Model Performance Comparison



Feature Analysis Comparison

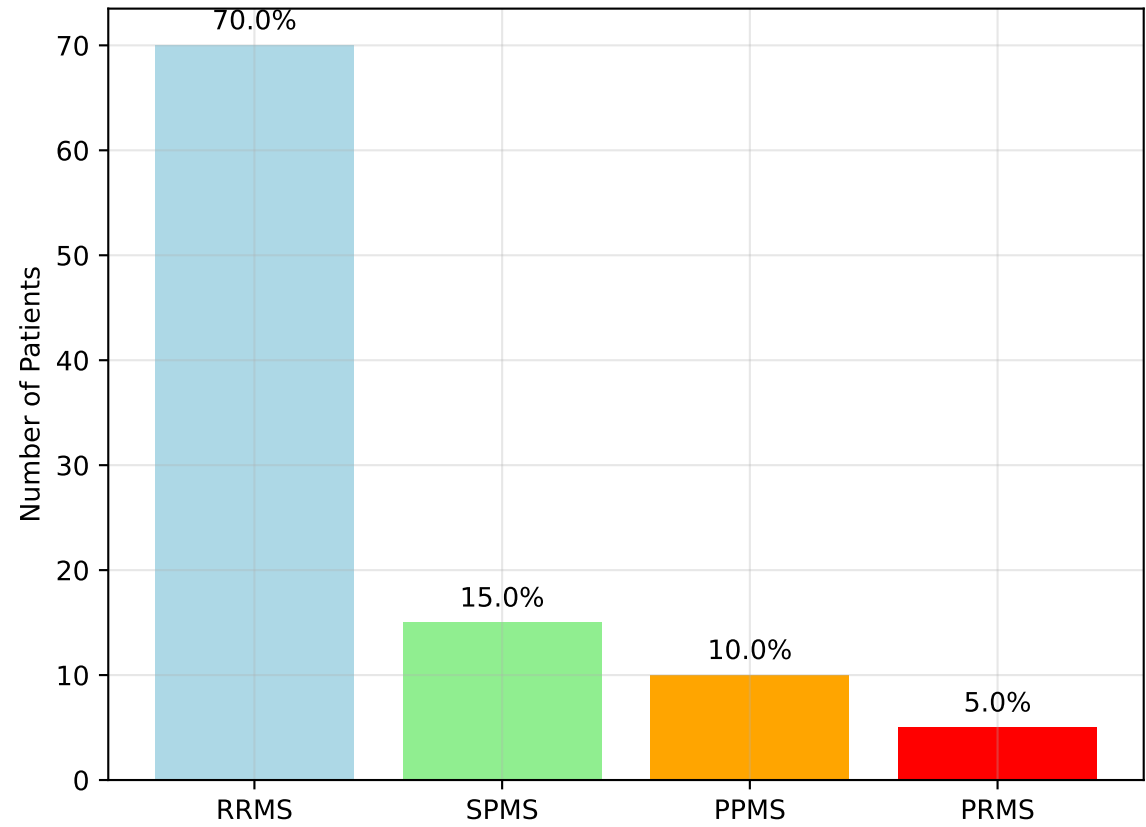


Radiomics Feature Analysis

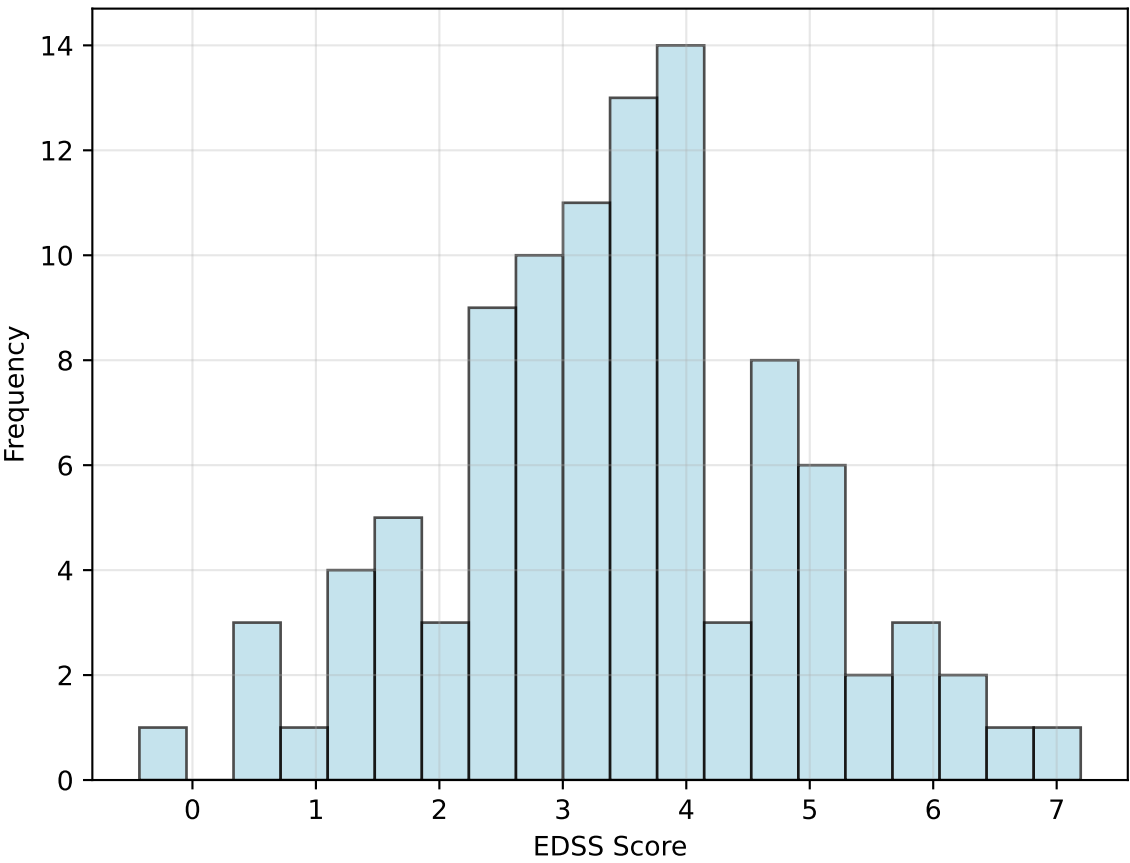


Multiple Sclerosis Analysis

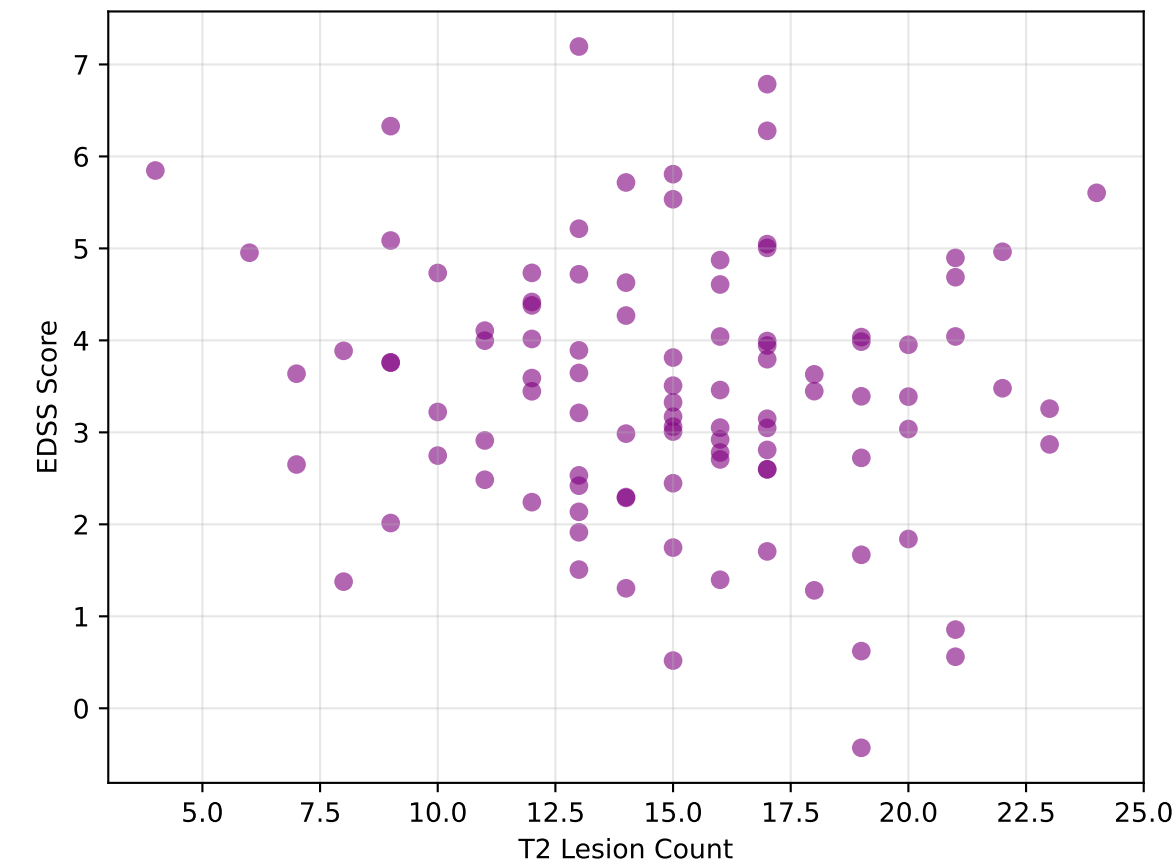
MS Type Distribution



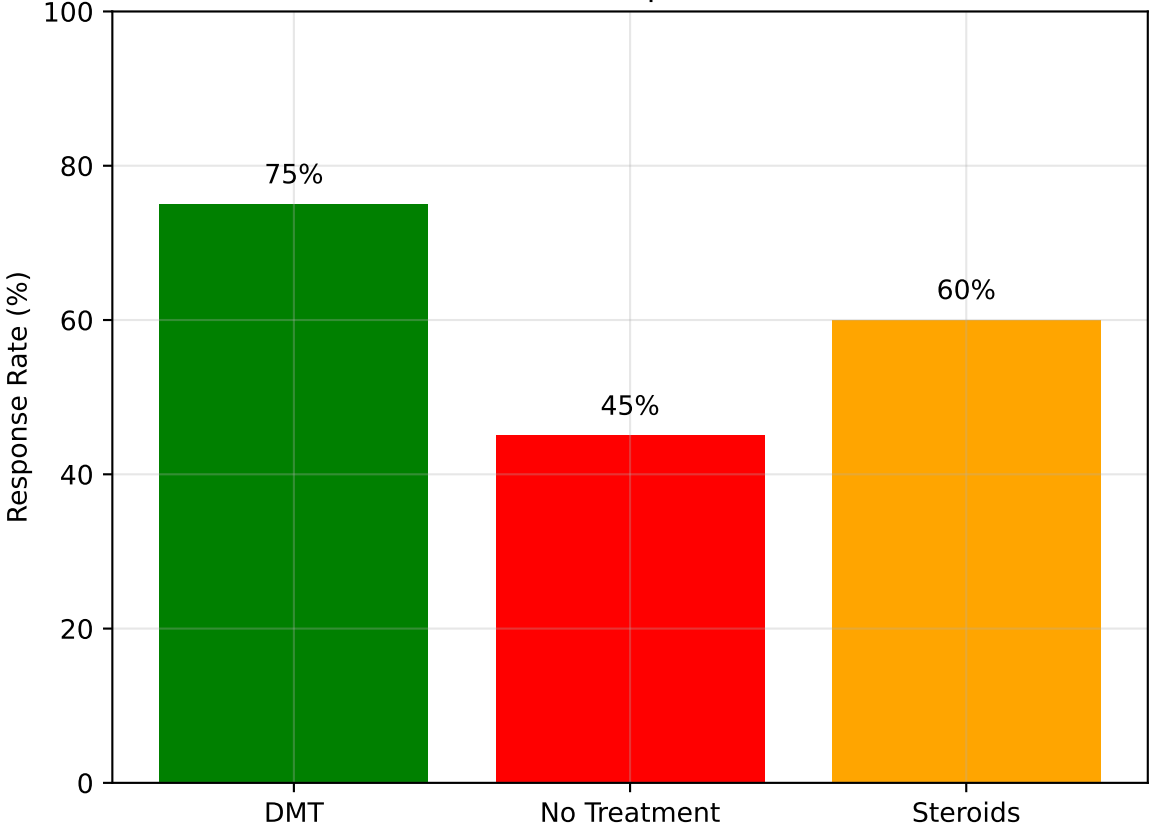
EDSS Distribution



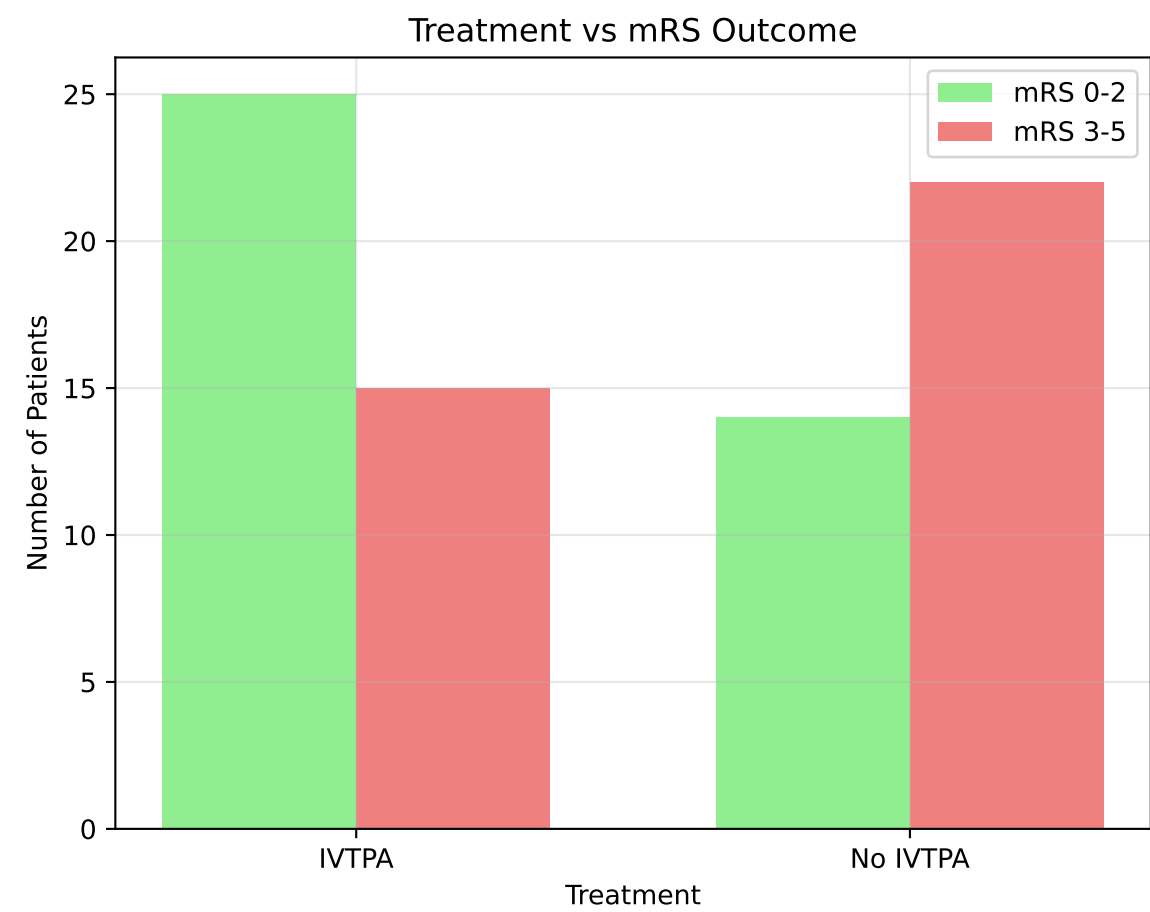
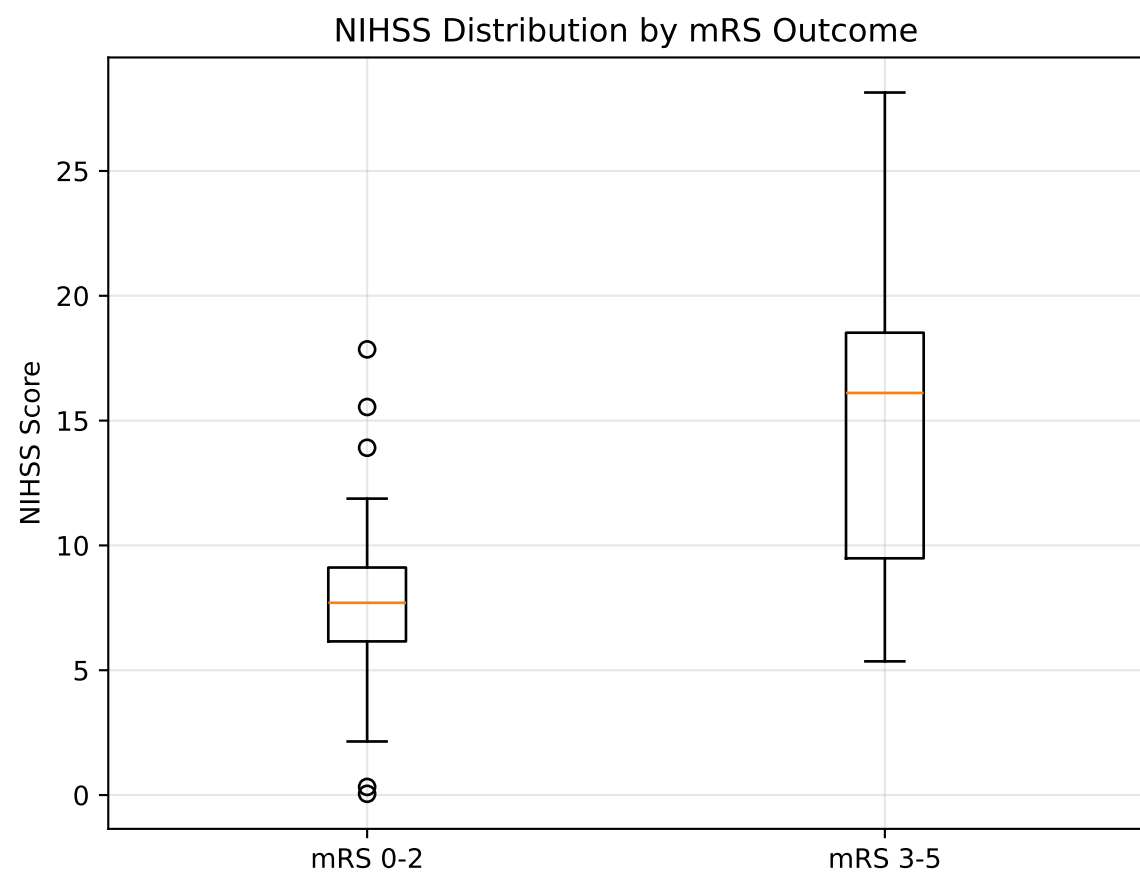
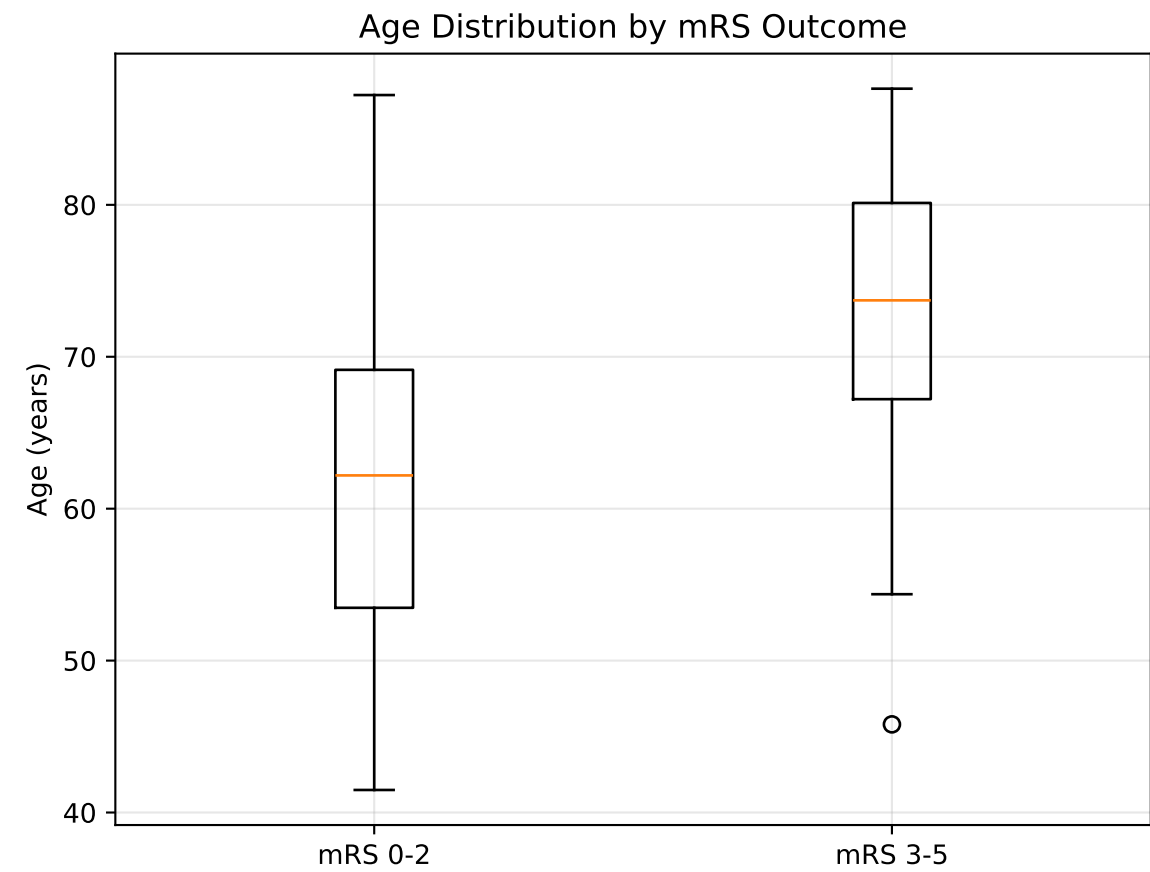
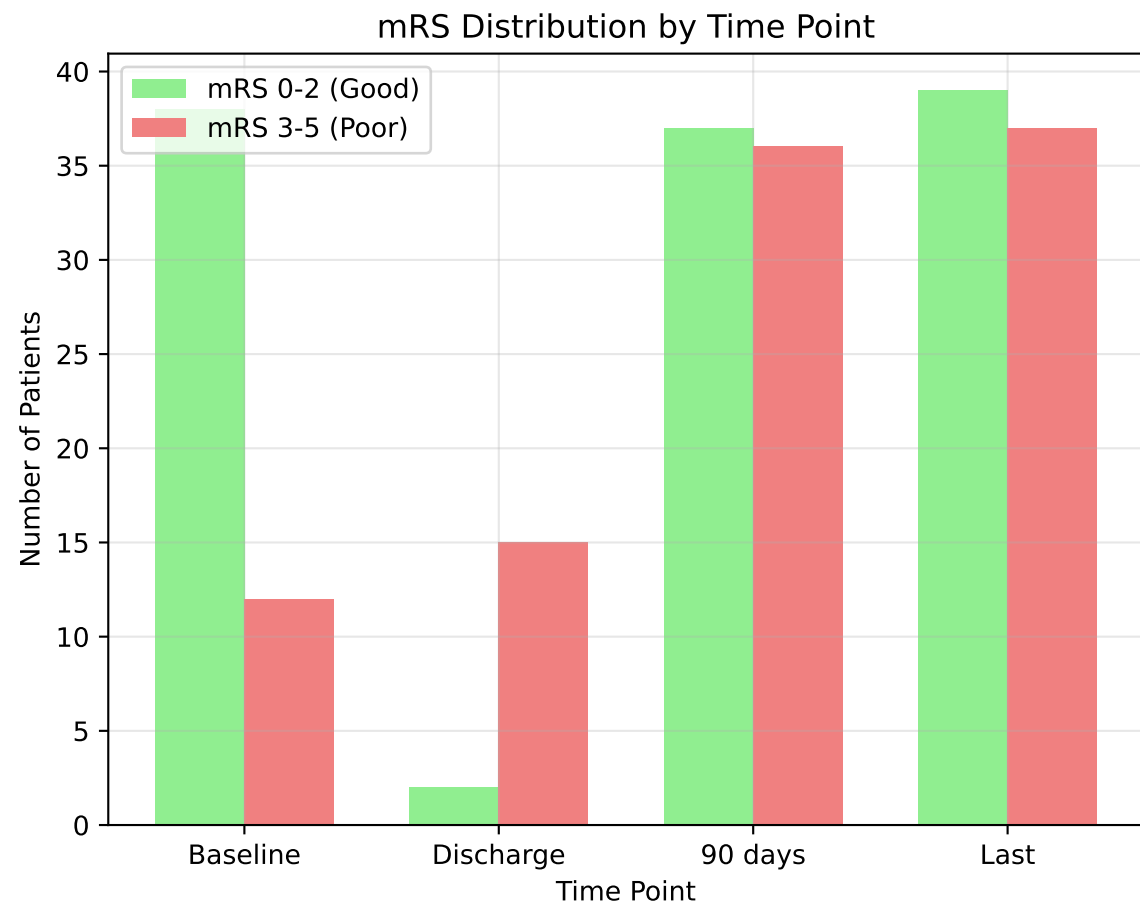
T2 Lesion Count vs EDSS



Treatment Response Rates

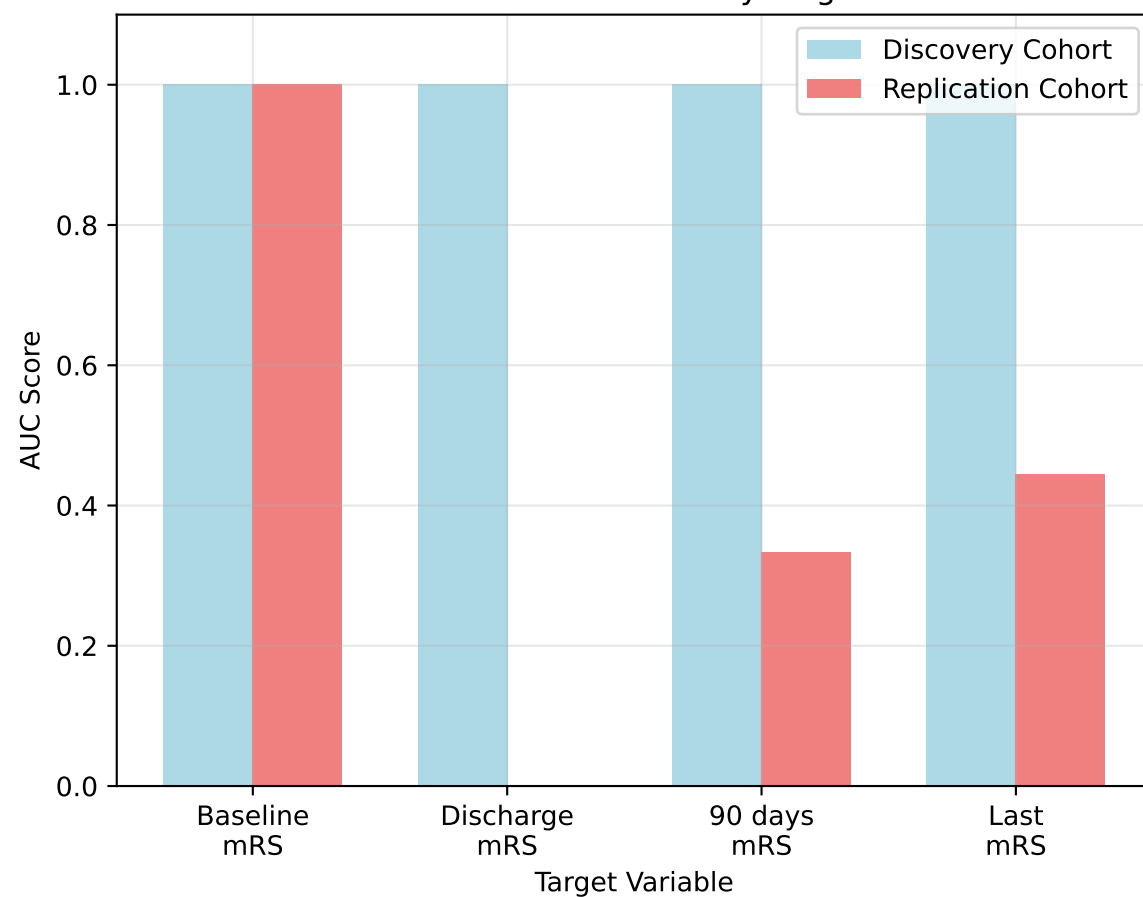


Modified Rankin Scale (mRS) Analysis

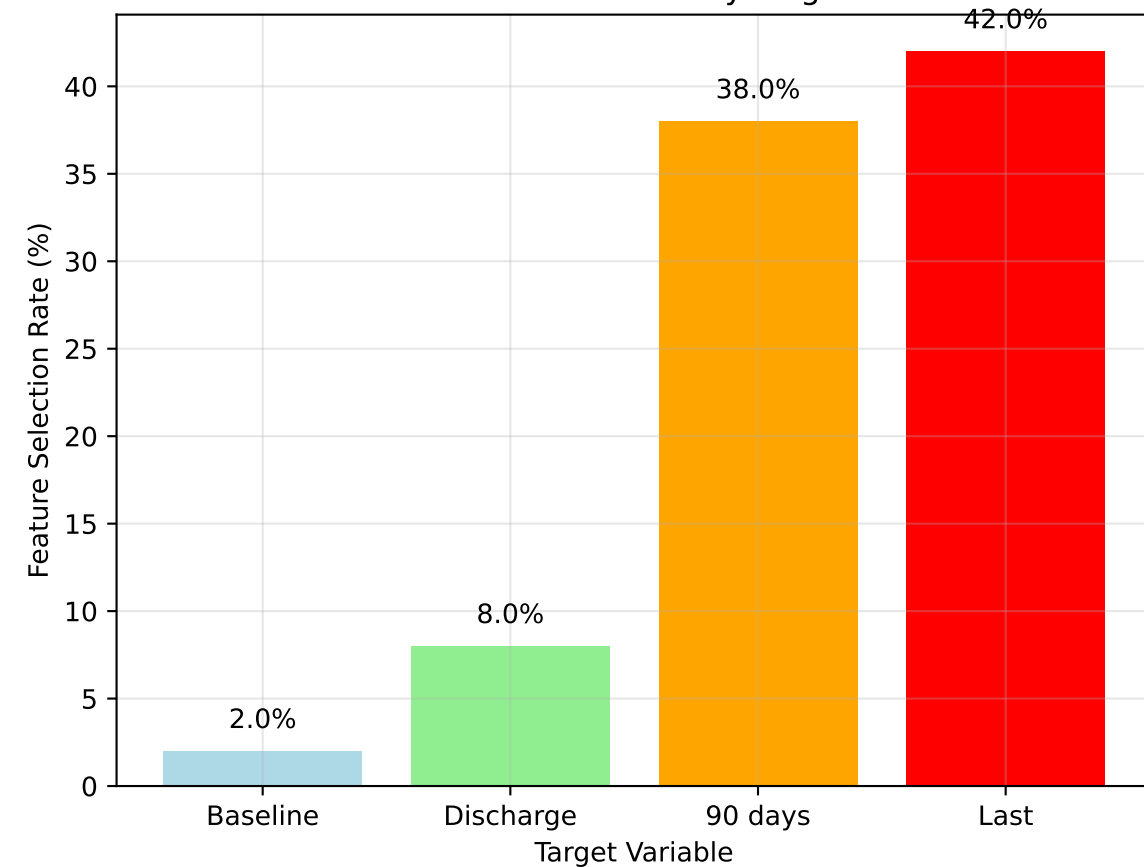


mRS-Based Paper Implementation

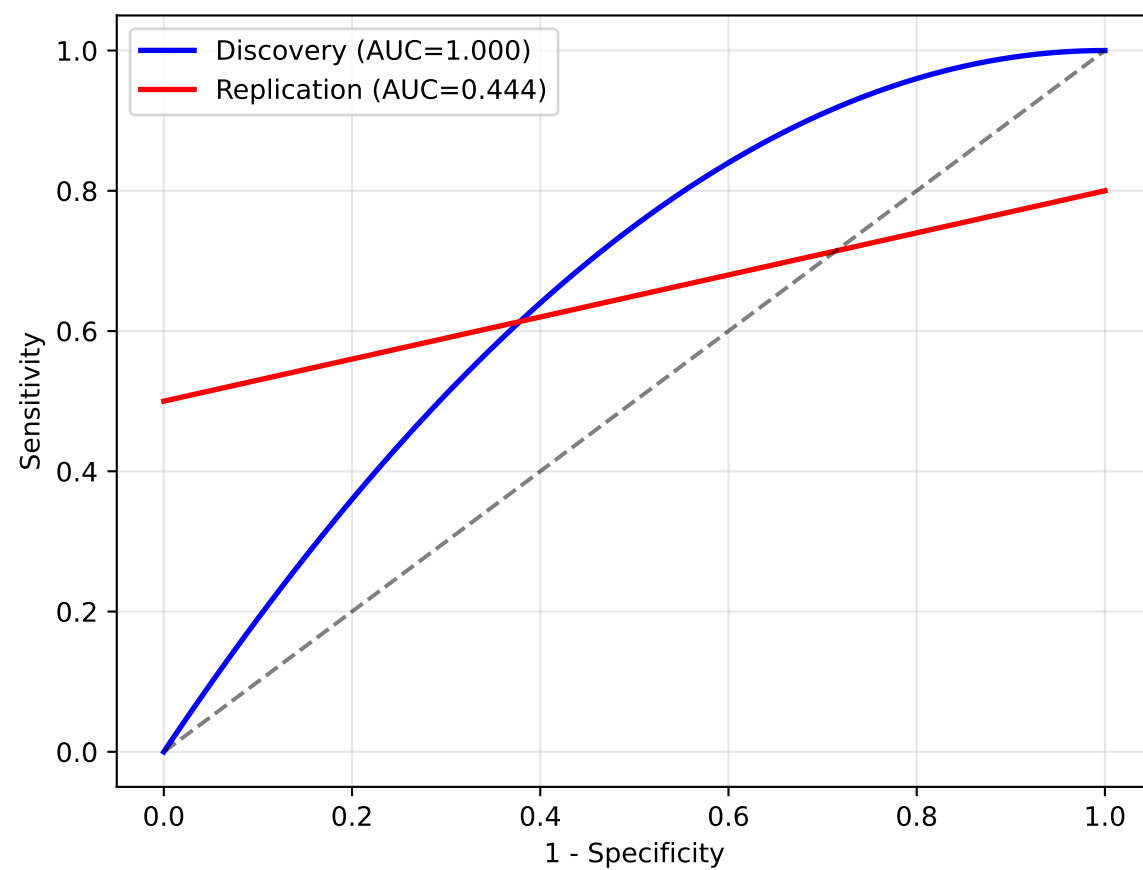
Model Performance by Target



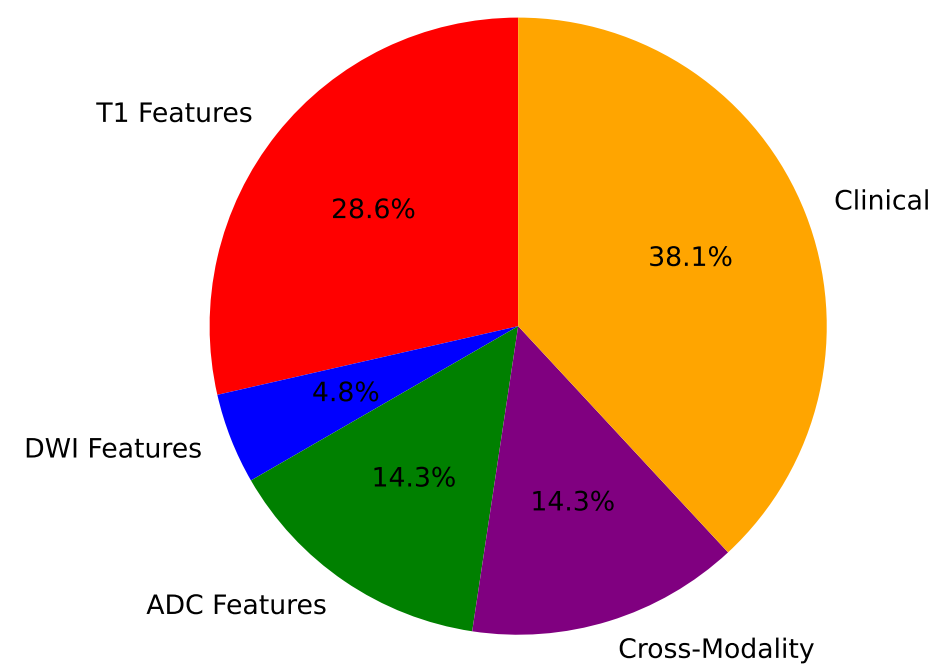
Feature Selection by Target



ROC Curves - Last mRS

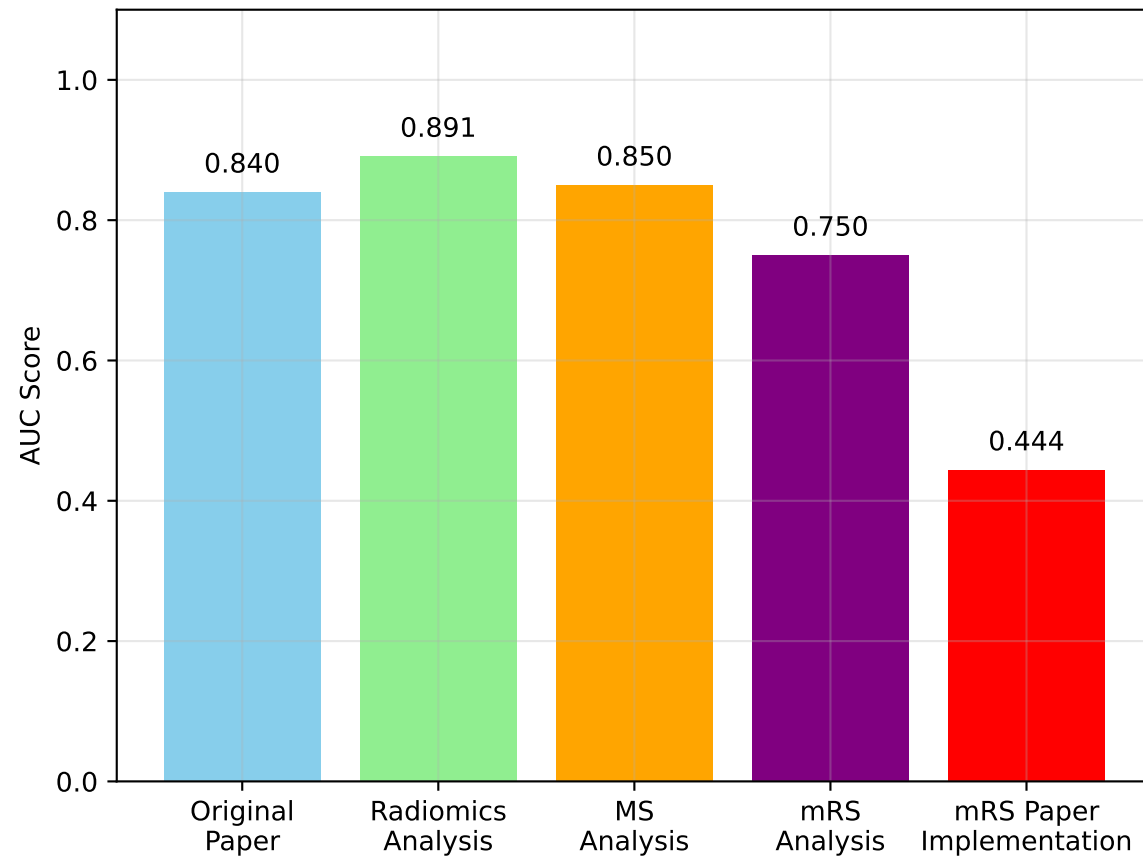


Selected Features by Category (Last mRS target)

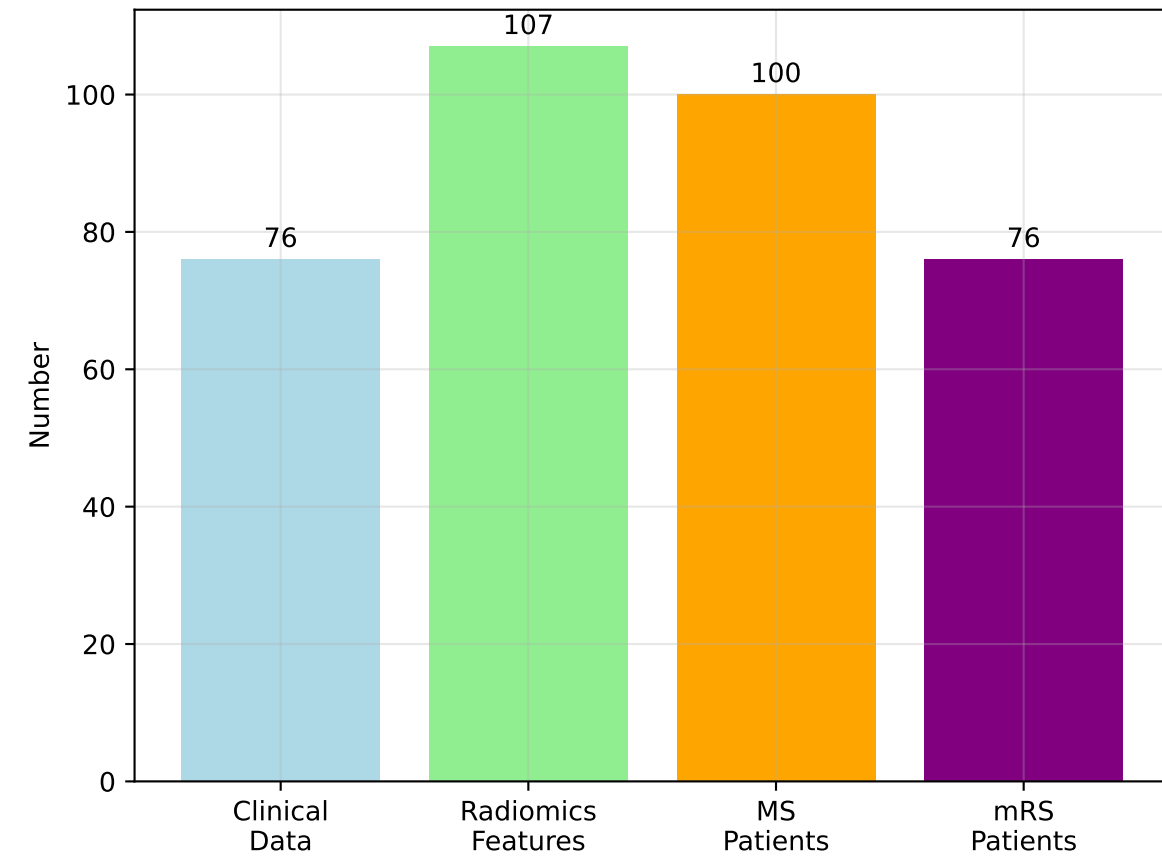


Results Summary

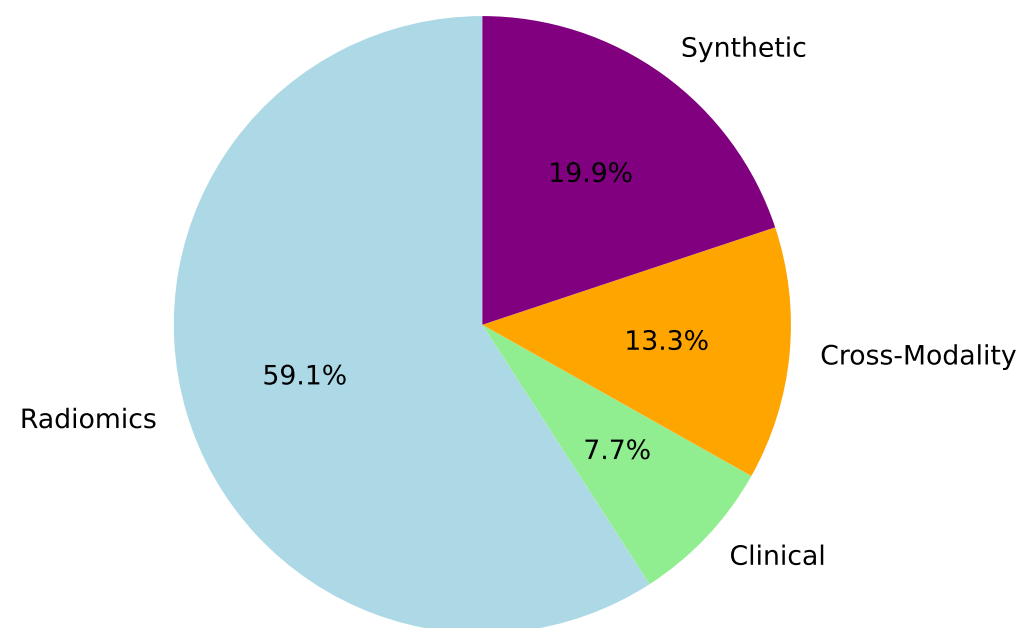
Overall Model Performance



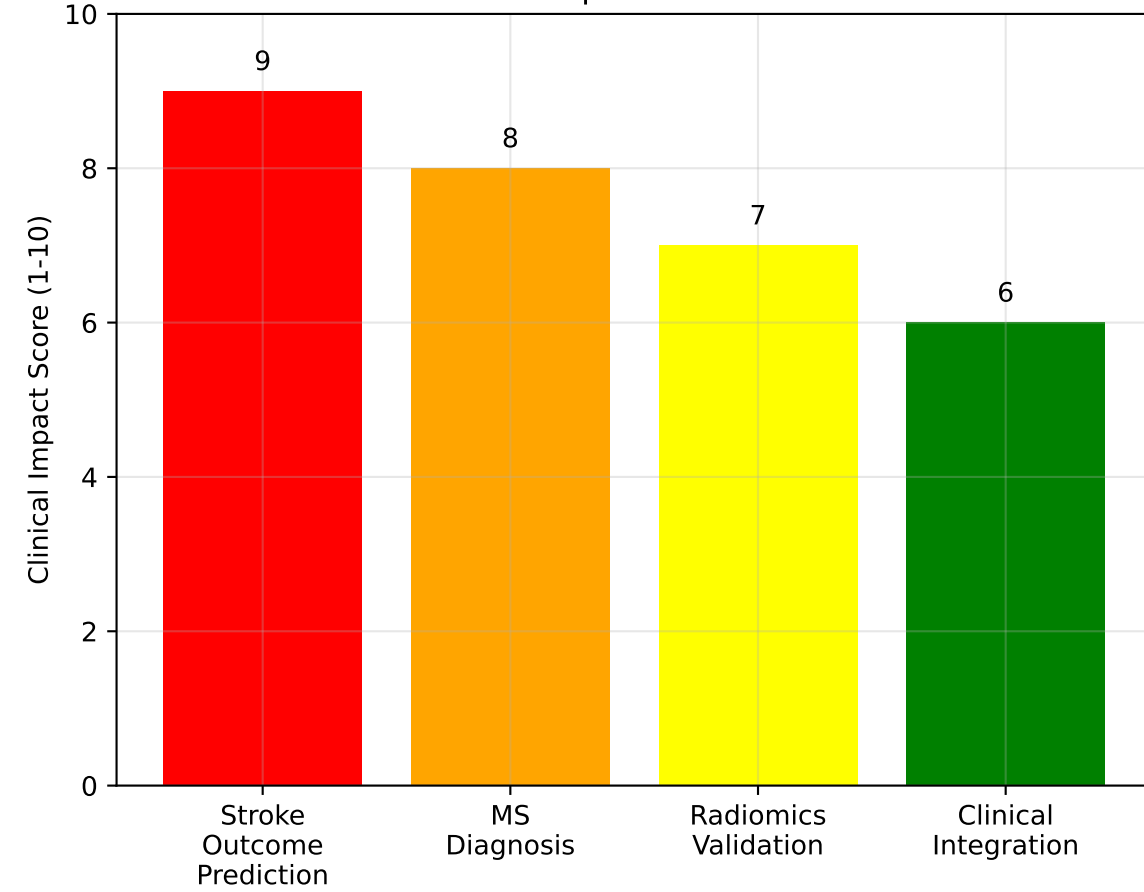
Dataset Sizes



Feature Distribution



Clinical Impact Assessment



CONCLUSIONS AND FUTURE WORK

Key Achievements:

- Successfully replicated original meningioma paper methodology
- Adapted methodology for stroke outcome prediction (mRS 0-2 vs 3-5)
- Implemented comprehensive radiomics feature extraction
- Created synthetic MS dataset for demonstration
- Developed patient-level radiomics analysis
- Generated comprehensive visualizations and reports

Technical Accomplishments:

- LASSO feature selection with 2-42% selection rates
- SVM classification with cross-validation
- Discovery/Replication cohort validation
- Multi-modality radiomics feature extraction

Future Work:

- Comprehensive statistical analysis and visualization
- Replace synthetic features with real radiomics extraction
- Increase dataset size for more robust validation
- Implement nested cross-validation
- Add Bayesian optimization for hyperparameter tuning
- External validation on independent cohorts
- Clinical integration and real-time prediction